```
clc;
clear;
close all;

%% ============================================================
% EXPERIMENT 5:
% PARKS-MCCLELLAN VS FREQUENCY-SAMPLING FIR FILTER DESIGN
%% ============================================================

%% 1. FILTER SPECIFICATIONS

Fs = 10000;         % Sampling frequency (Hz)
fp = 1500;          % Passband edge frequency (Hz)
fs = 2000;          % Stopband edge frequency (Hz)
N  = 40;            % Filter order

% Desired frequency response
% Passband gain = 1
% Stopband gain = 0

% Normalized frequency band edges (0 to 1)
f_bands = [0, fp, fs, Fs/2] / (Fs/2);

% Desired amplitudes
a_bands = [1, 1, 0, 0];

%% 2. PARKS-MCCLELLAN (EQUIRIPPLE) DESIGN

% Design FIR filter using Parks-McClellan algorithm
b_pm = firpm(N, f_bands, a_bands);

%% 3. FREQUENCY-SAMPLING DESIGN

% Design FIR filter using frequency-sampling method
b_fs = fir2(N, f_bands, a_bands);

%% 4. DISPLAY FILTER COEFFICIENTS

fprintf('\n============================================================\n');
```

```
============================================
```

```
fprintf(' EXPERIMENT 5: FIR FILTER DESIGN COMPARISON\n');
```

```
 EXPERIMENT 5: FIR FILTER DESIGN COMPARISON
```

```
fprintf('============================================================\n');
```

```
============================================
```

```
```

```
fprintf('\nFilter Order              = %d\n', N);
```

```
Filter Order              = 40
```

```
fprintf('Sampling Frequency       = %d Hz\n', Fs);
```

```
Sampling Frequency       = 10000 Hz
```

```
fprintf('Passband Edge Frequency  = %d Hz\n', fp);
```

```
Passband Edge Frequency  = 1500 Hz
```

```
fprintf('Stopband Edge Frequency  = %d Hz\n', fs);
```

```
Stopband Edge Frequency  = 2000 Hz
```

```

fprintf('\n--- PARKS-MCCLELLAN COEFFICIENTS (First 10) ---\n');
```

```
--- PARKS-MCCLELLAN COEFFICIENTS (First 10) ---
```

```
disp(b_pm(1:10));
```

```
   -0.0015    0.0072    0.0053   -0.0010   -0.0081   -0.0072    0.0037    0.0142    0.0100   -0.0090
```

```

fprintf('\n--- FREQUENCY-SAMPLING COEFFICIENTS (First 10) ---\n');
```

```
--- FREQUENCY-SAMPLING COEFFICIENTS (First 10) ---
```

```
disp(b_fs(1:10));
```

```
    0.0000    0.0001    0.0002   -0.0001   -0.0008   -0.0009    0.0008    0.0035    0.0029   -0.0039
```

```

%% 5. CALCULATE FREQUENCY RESPONSES

[H_pm, f] = freqz(b_pm, 1, 1024, Fs);
[H_fs, ~] = freqz(b_fs, 1, 1024, Fs);

% Magnitude response in dB
Mag_pm = 20*log10(abs(H_pm) + eps);
Mag_fs = 20*log10(abs(H_fs) + eps);

% Phase response in degrees
Phase_pm = unwrap(angle(H_pm)) * 180/pi;
Phase_fs = unwrap(angle(H_fs)) * 180/pi;

%% 6. MAGNITUDE RESPONSE COMPARISON

figure('Name', 'FIR Magnitude Response Comparison');

subplot(2,1,1);

plot(f, Mag_pm, 'b', 'LineWidth', 1.5);
```

```
hold on;
plot(f, Mag_fs, 'r--', 'LineWidth', 1.5);

grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');

title(['Magnitude Response Comparison (Order N = ', ...
       num2str(N), ')']);

legend('Parks-McClellan (Equiripple)', ...
       'Frequency-Sampling', ...
       'Location', 'best');

ylim([-80 10]);
xlim([0 Fs/2]);

hold off;

%% 7. PHASE RESPONSE COMPARISON

subplot(2,1,2);

plot(f, Phase_pm, 'b', 'LineWidth', 1.5);
hold on;
plot(f, Phase_fs, 'r--', 'LineWidth', 1.5);

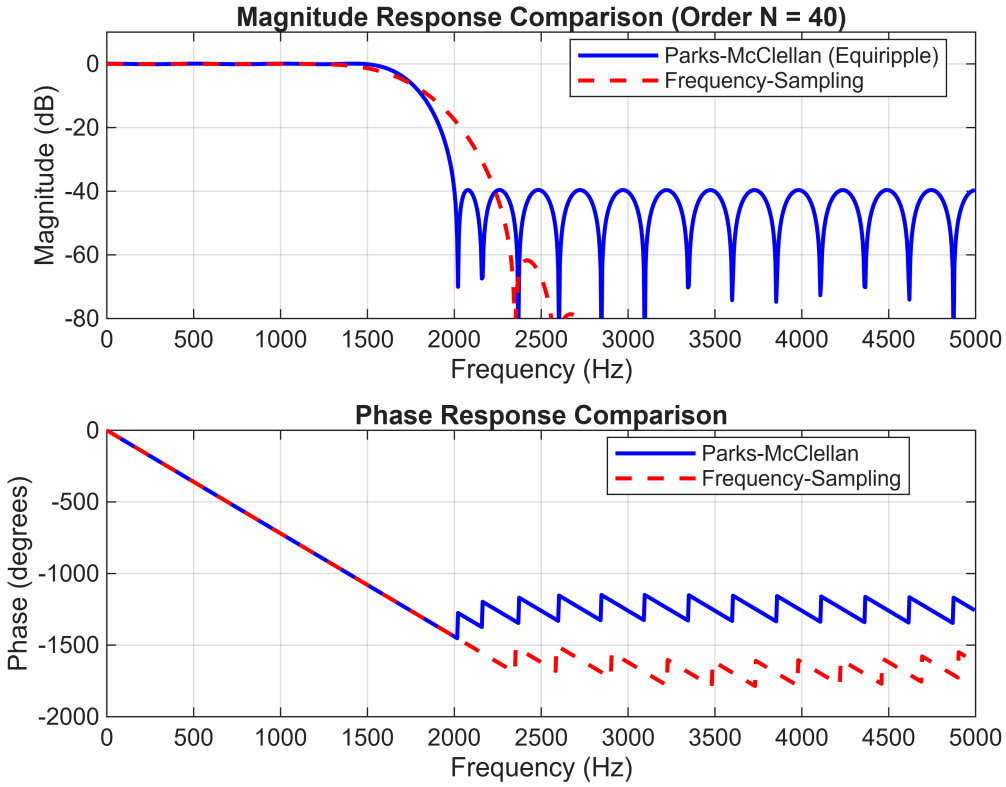
grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');

title('Phase Response Comparison');

legend('Parks-McClellan', ...
       'Frequency-Sampling', ...
       'Location', 'best');

xlim([0 Fs/2]);

hold off;
```




```
%% 8. END OF EXPERIMENT
```